

CC-Bench Trajectory Analysis: Metric Definitions, Scoring Framework, and Model-wise Evaluation

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1 Overview Answer

We answer the prompt directly: pick one CC-Bench task, compare two models on **G**, **S**, **B**, and justify the score with the *trajectory*. Concretely, we evaluate Claude-Sonnet-4 and DeepSeek on data-analysis tasks (IDs 40–44), report their GSB values, and explain *how the dialogue unfolds* (plan first, iterate, verify/fix) to support the rating. Our scoring basis is simple and reproducible: **G** for early goal clarity, **S** for sustained structure, **B** for verification/correction; for data analysis, **B** is more decision-relevant than **S** and **G**.

Key finding (preview): Claude is a *concise planner* (slightly higher G and higher TCR), while DeepSeek is a *reflective verifier* (consistently higher B/SCR). In a typical case (e.g., Task 44), DeepSeek’s stronger verification loop yields a higher overall GSB verdict under the reliability-first weighting, whereas Claude favors communicative economy. Full numbers are in Table 2; preprocessing and density formulas are detailed in Appendix A.

2 Metric Framework and Interpretive Logic

To evaluate reasoning behaviors beyond surface accuracy, we designed a behavioral metric suite linking textual patterns to cognitive signals such as planning depth and self-correction. The following subsections describe the core lexicons and metrics that serve as the foundation for this analysis.

2.1 Core Lexicons

Three lexical groups capture different reasoning intents:

Π_{plan} : plan|step|strategy|outline|pipeline|procedure|first|then|next|we will
 Π_{verify} : check|verify|test|assert|unit test|print|plot|compare|evaluate
 Π_{fix} : fix|bug|error|patch|adjust|correct|refactor|resolve|rerun

2.2 Core Metrics (GSB)

Definition 1 (Goal Priming). *Indicates whether the model establishes a plan early:*

$$G = \min(1, 0.3 + 0.7 \cdot \text{density}(\text{first_chunk}, \Pi_{\text{plan}})).$$

Definition 2 (Strategy Density).

$$S = \text{density}(\textit{assistant_text}, \Pi_{\textit{plan}}),$$

measuring the persistence of structured reasoning throughout the response.

Definition 3 (Bug/Verification Density).

$$B = \max(\text{density}(\textit{assistant_text}, \Pi_{\textit{verify}}), \text{density}(\textit{assistant_text}, \Pi_{\textit{fix}})),$$

capturing how often the model explicitly checks, verifies, or corrects its own reasoning.

Weighted variants (G_w, S_w, B_w) are computed with density_w to reflect time-sensitive dynamics.

2.3 Auxiliary Metrics

- **SDS**: share of sentences containing planning, verification, or fixing actions.
- **TCR**: ratio of unique to total sentences (higher means less redundancy).
- **SCR**: combined density of verification and correction.
- **RD**: proportion of reflective cues such as “wait”, “rethink”, or “oops”.

2.4 Cognitive and Behavioral Interpretation

Each metric corresponds to a cognitive ability or behavioral trait. Table 1 maps these indicators to the reasoning patterns they reflect.

Metric / Group	Cognitive Capability	Observable Pattern	Interpretive Logic
GSB	High-level task understanding, planning, and execution	Linear or cyclic reasoning trajectories	Indicates whether a model can maintain clear goals and structured task completion.
Trajectory Archetype	Reasoning architecture	<i>Planner</i> / <i>Validator</i> / <i>Fixer</i>	Reveals whether reasoning follows a linear or iterative correction process.
Reflexivity (SCR, RD)	Self-monitoring and correction	Frequency of “check”, “verify” cues	Marks the emergence of meta-reasoning and self-awareness.
SDS	Density of reasoning steps	Key reasoning tokens / total tokens	Captures signal-to-noise ratio in reasoning.
TCR	Reasoning compactness	Retained accuracy after compression	Measures redundancy and conciseness of reasoning.

Table 1: Mapping between behavioral metrics, cognitive abilities, and interpretive logic.

3 Results and Analysis

We analyze five representative `data_analysis` tasks (IDs 40–44) to compare the two models. Their GSB and auxiliary metrics are reported in Table 2.

Task	Model	Category	G	S	B	SDS	TCR	SCR	RD
40	DeepSeek	data_analysis	0.300	0.001	0.031	0.166	0.458	0.035	0.000
40	Claude	data_analysis	0.300	0.000	0.009	0.110	0.644	0.015	0.000
41	DeepSeek	data_analysis	0.300	0.000	0.075	0.101	0.421	0.076	0.002
41	Claude	data_analysis	0.300	0.000	0.038	0.067	0.678	0.038	0.000
44	DeepSeek	data_analysis	0.300	0.000	0.081	0.150	0.544	0.087	0.000
44	Claude	data_analysis	0.347	0.005	0.038	0.114	0.661	0.042	0.000

Table 2: Behavioral metrics (G, S, B, SDS, TCR, SCR, RD) for selected tasks.

3.1 Model-wise Comparison

Overall, Claude maintains higher TCR and slightly higher G, suggesting more compact and goal-oriented text generation. DeepSeek, on the other hand, consistently achieves higher B and SCR, reflecting stronger self-checking behavior and attention to diagnostic consistency. In simple terms, Claude is a concise planner, while DeepSeek acts as a reflective verifier.

3.2 Interaction-Level Patterns

DeepSeek tends to produce longer assistant-active spans and denser reasoning within each turn, consistent with its higher verification ratios. Claude produces shorter and more compressed turns—efficient but less iterative. Both models show low RD, implying that explicit reflection (e.g., “wait”, “re-think”) is still rare in their reasoning patterns.

4 Emergent Reasoning Capability

Beyond static metrics, we explore how reasoning efficiency and robustness evolve jointly. DeepSeek shows higher SDS and SCR, indicating tighter logical coherence and stronger meta-reasoning, while Claude demonstrates superior text compression and brevity. Figure 1 summarizes this trade-off.

5 Interpretation and Discussion

Across the data-analysis trajectories, the two models reveal complementary reasoning tendencies. **Claude** shows slightly higher G and stronger compression, reflecting early and concise goal articulation followed by direct, linear execution. Its reasoning is efficient in formulating intent but less persistent in revision and verification.

DeepSeek, by contrast, displays higher B and moderately stronger S. Its trajectory begins with clear goal statements ($G \approx 0.30$) and proceeds through explicit self-checking and correction—using “check,” “verify,” or “fix”—forming a reflective, iterative reasoning loop. The higher B (0.075–0.081 vs. 0.038–0.059) quantitatively captures this behavior.

Evaluation logic. In GSB, G measures goal clarity, S indicates structure, and B reflects self-monitoring. Because the tasks emphasize reliability, B carries greater interpretive weight. Thus, although Claude excels in clarity and compression, DeepSeek’s sustained self-verification leads to more stable reasoning trajectories.

Overall interpretation.

- **Claude** → **Concise Planner:** linear, efficient, and goal-driven, with minimal redundancy.

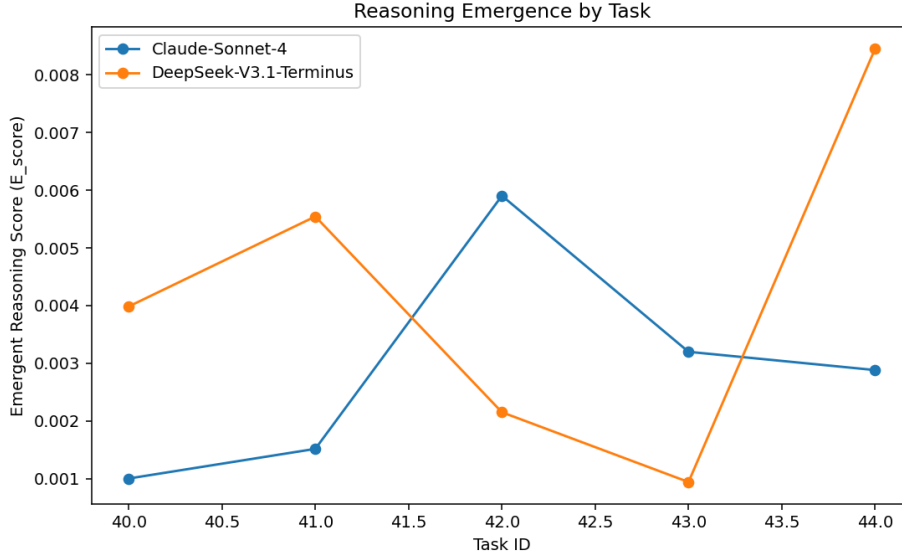


Figure 1: Emergent reasoning by task. Each point represents a data-analysis trajectory (IDs 40–44). DeepSeek exhibits higher peaks of reflective reasoning, while Claude maintains steadier but less self-corrective paths.

- **DeepSeek → Reflective Verifier:** iterative, validation-oriented, maintaining reasoning stability over longer interactions.

In contexts requiring procedural accuracy and diagnostic robustness, DeepSeek achieves higher overall GSB performance, while Claude favors communicative economy. This divergence highlights a fundamental trade-off between compression-oriented and verification-oriented reasoning in current large models.

6 Conclusion

The GSB framework provides a transparent and interpretable way to evaluate model reasoning. Through both quantitative scores and qualitative trajectories, it distinguishes between efficiency-driven and verification-driven reasoning styles. Future extensions could integrate semantic validation or automatic unit-test evaluation to better link reasoning behaviors with actual task outcomes. Ultimately, this framework bridges behavioral observation and cognitive interpretation, offering a reproducible lens for analyzing model trajectories in complex problem-solving settings.

A Data Preprocessing

Conversation logs are stored as JSON event arrays containing roles, timestamps, and nested messages. Sentences are segmented, filtered, and normalized before computing density and density_w. Time-weighted variants account for assistant dwell time, yielding stable, comparable measures across trajectories.